GTSAM

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Project Overview

This project focuses on contributing to the Georgia Tech Smoothing and Mapping (GTSAM) library, an open-source C++ library that implements smoothing and mapping (SAM) in robotics and vision using factor graphs and Bayes networks as the underlying computing paradigm.

Our work over the summer targeted improving GTSAM's documentation, examples, and control-related extensions to make it more accessible to researchers and practitioners.

Goals and Milestones (Oliver)

Goals:

- Enhance GTSAM's documentation for classes in the geometry module.
- Provide clear, interactive Python examples to support learning and research.

Milestones:

- Authored user guides for all major camera calibration models and stereo components.
- Ported and improved StereoVO examples with interactive Plotly visualization.
- Investigated Python/C++ wrapper internals

Goals and Milestones (Zhouyu)

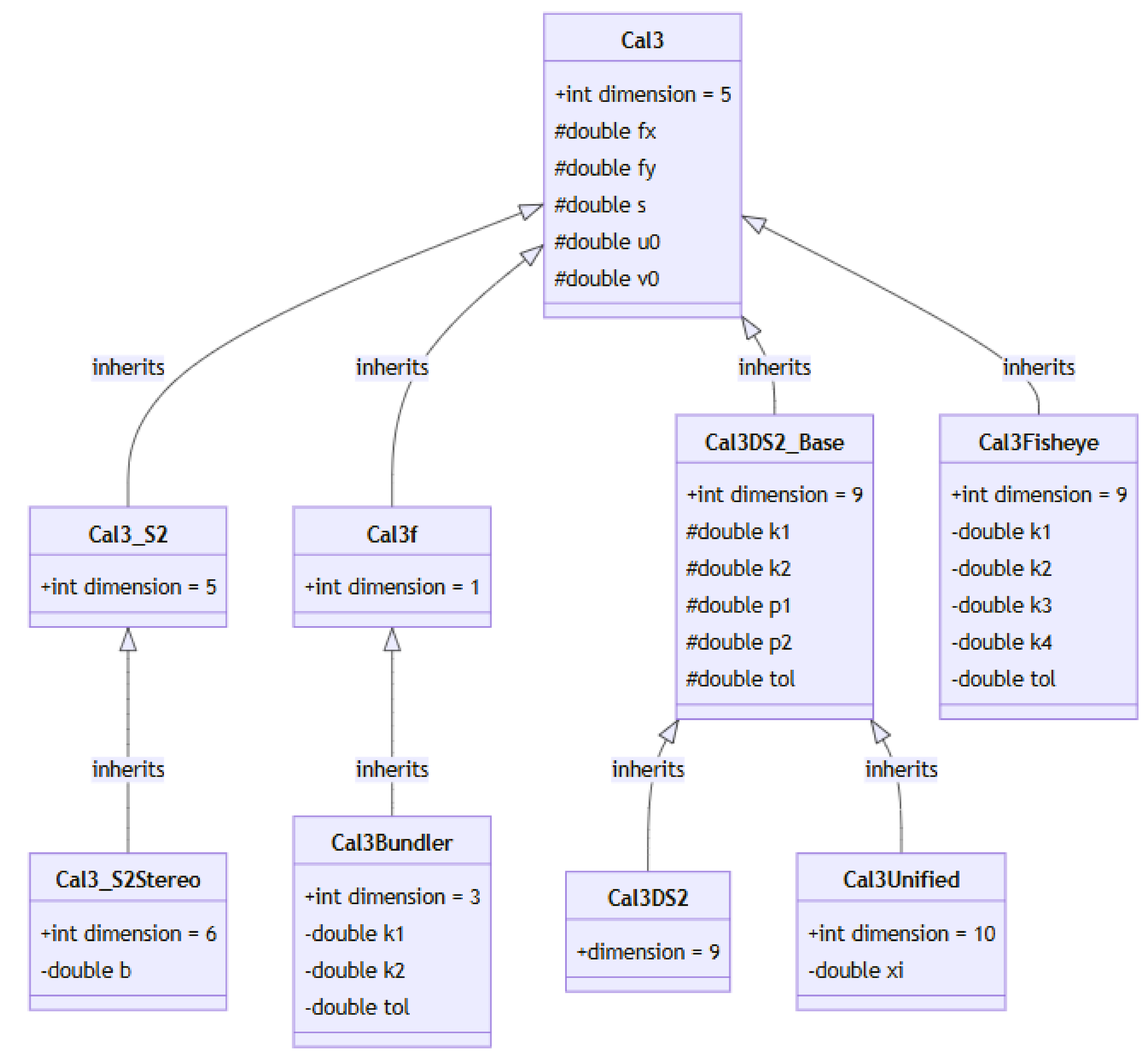
Goals:

- Enrich GTSAM's python examples
- Write GTSAM use cases that link optimal control and graph optimization.

Milestones:

- Composed examples that implements linear quadratic regulator (LQR) in GTSAM for linear systems
- Authored iterative LQR (iLQR) examples for nonlinear systems
- Created example using GTSAM LQR for an actual control problem: inverted pendulum

Highlights and Accomplishments (Oliver)



- Authored comprehensive user guides for all major calibration models and stereo vision components.
- Ported and enhanced Stereo Visual Odometry (StereoVO) examples from C++ to Python notebooks, including an interactive Plotly visualization for result analysis.
- Explored the Python wrapper internals for GTSAM and fixed many wrapper errors.
- Worked on fixing existing user guides and documentation pages
  - Created visual components like UML diagrams to improve conceptual clarity for new users.
  - Fixed broken links.
  - Worked on standardizing all the formatting of the documentation pages.
  - Provided cross-linking between guides to allow easy navigation to related topics.
  - Added additional resources section to the user guides to provide more in-depth theoretical information.

Highlights and Accomplishments (Zhouyu)

$$x_{t+1} = Ax_t + Bu_t, \quad x_0 \text{ given}$$

$$\min_{\{u_t\}_{t=0}^{T-1}} \quad J = \sum_{t=0}^{T-1} \left(x_t^\top Q x_t + u_t^\top R u_t\right) + x_T^\top Q_f x_T$$

$$\text{s.t.} \quad x_{t+1} = Ax_t + Bu_t$$

LQR problem statement

$$x_{t+1} = f(x_t, u_t), \quad x_0 \text{ given}$$

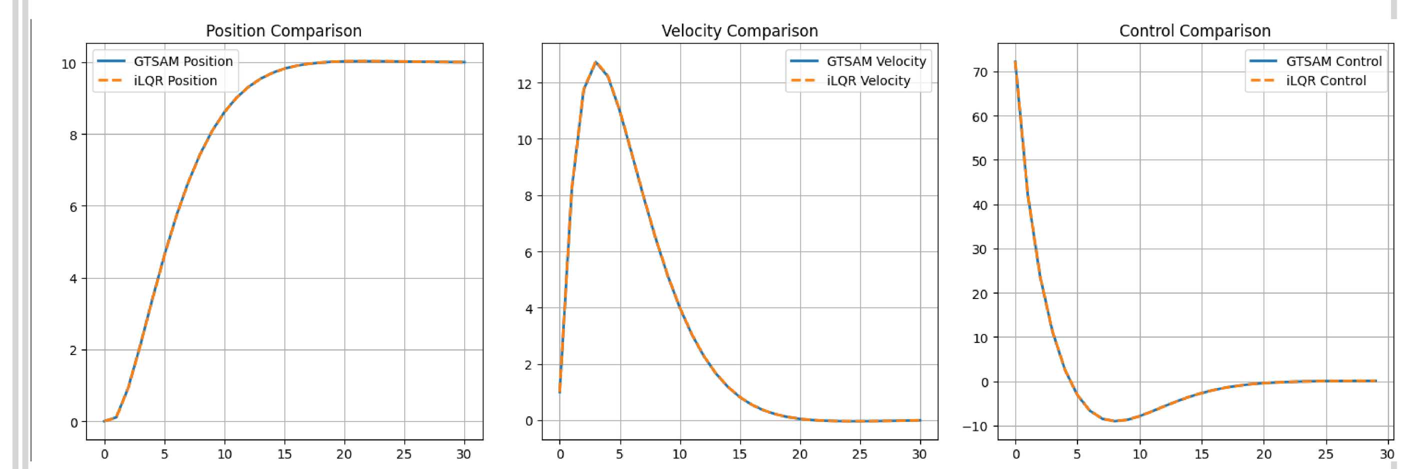
$$\min_{\{u_t\}_{t=0}^{T-1}} \quad J = \sum_{t=0}^{T-1} \ell(x_t, u_t) + \ell_T(x_T)$$

$$\text{s.t.} \quad x_{t+1} = f(x_t, u_t)$$

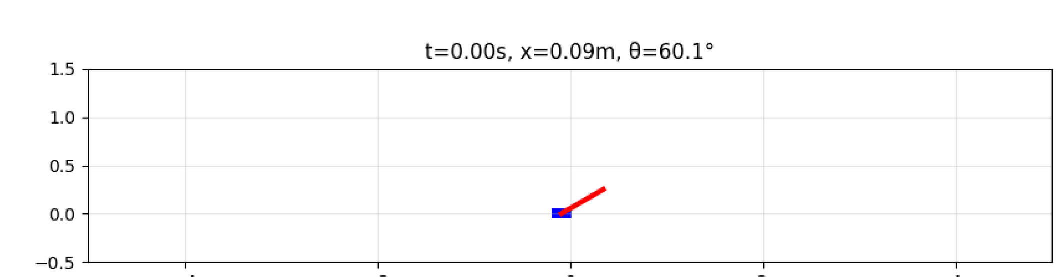
iLQR problem statement

Nonlinear damping system

$$f(x_t, u_t) = \begin{bmatrix} \text{pos}_{t+1} \\ \text{vel}_{t+1} \end{bmatrix} = \begin{bmatrix} \text{pos}_t + \text{vel}_t \cdot \Delta t \\ \text{vel}_t + (u_t - 0.1 \cdot \text{vel}_t^2) \cdot \Delta t \end{bmatrix}$$



- **LQR**
- Built a factor-graph version of the Linear Quadratic Regulator (LQR) for linear dynamics.
- Used GTSAM to solve the discrete-time optimal control problem by constructing factors that represent system dynamics and quadratic costs.
- **iLQR**
- Authored an example that applies iterative LQR using nonlinear dynamics.
- Used GTSAM's nonlinear factor framework to manage repeated optimization and feedback computation.
- Demonstrated that solution from GTSAM naturally mirrors the vanilla method (for both LQR and iLQR).



Open Source Outcomes

(Oliver and Zhouyu)

- Gained firsthand experience of the opensource contribution pipeline
- Got much better working with Git CLI and Github
- Learned how to work on forked repository and PRs

Future Work

(Oliver)

- Continue contributing to GTSAM documentation and user guides
- Added more interesting and visual examples
- Compose articles on theoretical foundation

(Zhouyu)

- Continue contributing to GTSAM documentation and user guides
- Explore inter-discinplinary research opportunities

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